



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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**Class : 11**

UNIT TEST-2, AUGUST - 2024

BUSINESS MATHEMATICS AND STATISTICS

Time Allowed : 1.30 Hours

[Max. Marks : 45]

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark.

10x1=10

1. The x-intercept of the straight line $3x + 2y - 1 = 0$ is
(1) 3 (2) 2 (3) $\frac{1}{3}$ (4) $\frac{1}{2}$
2. If $kx^2 + 3xy - 2y^2 = 0$ represent a pair of lines which are perpendicular then k is equal to
(1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) 2 (4) -2
3. The centre of the circle $x^2 + y^2 - 2x + 2y - 9 = 0$ is
(1) (1,1) (2) (-1, -1) (3) (-1,1) (4) (1, -1)
4. The eccentricity of the parabola is
(1) 3 (2) 2 (3) 0 (4) 1
5. The angle between the line $y = x$ and $y = -x$ is
(1) 45° (2) 90° (3) 180° (4) 60°
6. The value of $\sin 15^\circ$ is
(1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (2) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (3) $\frac{\sqrt{3}}{\sqrt{2}}$ (4) $\frac{\sqrt{3}}{2\sqrt{2}}$
7. If $\sin A + \cos A = 1$ then $\sin 2A$ is equal to
(1) 1 (2) 2 (3) 0 (4) $\frac{1}{2}$
8. $\tan\left(\frac{\pi}{4} - x\right)$ is
(1) $\left(\frac{1+\tan x}{1-\tan x}\right)$ (2) $\left(\frac{1-\tan x}{1+\tan x}\right)$ (3) $1 - \tan x$ (4) $1 + \tan x$
9. If $\operatorname{psec} 50^\circ = \tan 50^\circ$ then p is
(1) $\cos 50^\circ$ (2) $\sin 50^\circ$ (3) $\tan 50^\circ$ (4) $\sec 50^\circ$
10. Value of $\sec^2 63^\circ - \tan^2 63^\circ$ is
(1) 0 (2) 1 (3) -1 (4) ∞

PART - II

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question 16 is compulsory

4x2=8

11. Find the values of : $\sin 150^\circ$.
12. Find the equation of the circle having (4,7) and (-2,5) as the extremities of a diameter.
13. Find the length of the tangent from (1,2) to the circle $x^2 + y^2 - 2x + 4y + 9 = 0$.
14. Convert the following into the product of trigonometric functions: $\sin 7\theta - \sin 4\theta$.
15. Find the angle between the pair of straight lines $3x^2 - 5xy - 2y^2 + 17x + y + 10 = 0$.
16. Find the value of $\tan 15^\circ$.

PART - III

1. Answer any 4 questions
2. Each question carries 3 marks
3. Question 22 is compulsory

4 × 3 = 12

17. Find the value of P if the line $3x + 4y - P = 0$ is a tangent to the circle $x^2 + y^2 = 16$.
18. Using multiple angle identity, find $\tan 60^\circ$.
19. Solve $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{4}{7}\right)$.
20. If A, B, C, D are angles of a cyclic quadrilateral, prove that : $\cos A + \cos B + \cos C + \cos D = 0$.
21. Find the value of 'a' for which the straight lines $3x + 4y = 13$; $2x - 7y = -1$ and $ax - y - 14 = 0$ are concurrent.
22. Find the vertex, Axis, Length of Latus rectum of the parabola $(x-1)^2 = 8(y-2)$.

PART - IV

1. Answer all the questions
2. Each question carries 5 marks

3 × 5 = 15

23. a) If the distance of a point from the points (2,1) and (1,2) are in the ratio 2:1, then find the locus of the point.

(OR)

- b) Find the equation of the circle passing through the points (0,0), (1,2) and (2,0).

24. a) If $\tan \alpha = \frac{1}{3}$ and $\tan \beta = \frac{1}{7}$ then prove that $(2\alpha + \beta) = \frac{\pi}{4}$.

(OR)

- b) Prove that $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$.

25. a) Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.

(OR)

- b) A private company appointed a clerk in the year 2012, his salary was fixed as ₹ 20,000. In 2017 his salary raised to ₹ 25,000. (i) Express the above information as a linear function in x and y where y represent the salary of the clerk and x-represent the year

(ii) What will be his salary in 2020?



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